**LESSON 120 (0.5 GROUND and 1.5 DUAL)**

**Lesson Objective**

During this lesson, the applicant will gain proficiency with emergency procedures (in the pattern), traffic pattern procedures, and ground reference maneuvers. The applicant will be introduced to the no flap and power off approach as well as the rejected takeoff.

**Briefing**

- Review homework assigned in the previous lesson.
  - Quiz the student on emergency procedures and checklists.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

**References**

- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
  - Chapter 05 - Aerodynamics of Flight.
  - Chapter 12 - Weather Theory.
- Aeronautical Information Manual. (AIM)
  - Chapter 06 - Emergency Procedures.
  - Chapter 07 - Safety of Flight.
    - Section 04 - Wake Turbulence.
- Airplane Flying Handbook. (FAA-H-8083-3C)
  - Chapter 08 - Airport Traffic Patterns.
  - Chapter 09 - Approaches and Landings.

**Instructor Guidance**

- This is the last dual lesson before the pre-solo tests in Lesson 121 — your honest assessment of the student's readiness today directly determines whether they proceed. If there are meaningful gaps in any area, communicate them clearly in the post-brief and set specific expectations for what must be improved before Lesson 121.
- Quiz the student on emergency checklists and light gun signals during the ground portion. Both are tested in Lesson 121 and must be memorized — not just reviewed. If the student cannot recite them accurately, delay the flight and address the gaps on the ground first.
- For the rejected takeoff and engine failure after takeoff, brief each scenario clearly before departure. The rejected takeoff should be conducted with ATC coordination — ensure the student executes the correct procedure without prompting. The engine failure after takeoff is a talk-through exercise only, not a physical execution.
- Normal landings in this lesson must be completed without instructor guidance per the completion standard. Keep your hands in your lap unless a safety intervention is required. If you cannot do that, the student is not ready for the pre-solo tests and that conversation needs to happen in the post-brief.
- Thoroughly brief the no-flap and power-off approaches before the flight — both produce a significantly different sight picture and energy state than a normal approach. A student who

encounters either one unprepared will struggle. If the power-off approach cannot be landed in the first third of the runway, execute a go-around without hesitation.

**Lesson Content (Ground)**

**Radio Communications Failure Procedures**

- The first step is always to troubleshoot:
  - Double check the volume.
    - Audio panel.
    - Comm volume.
    - Headset volume.
  - Check to make sure the headset is plugged in.
  - Ensure the frequency is set correctly.
  - Attempt using the alternate communications radio (com1/com2).
  - Attempt to call the tower with a cell phone.
- If troubleshooting is unsuccessful, squawk 7600.
- If close to a non-towered airport, proceed there.
- If not, circle over the towered airport at 500' above traffic pattern altitude.
- Review light gun signals.

**Rejected Takeoff**

- A rejected takeoff can be due to many factors, such as:
  - Engine failure.
  - High oil pressure/temperature.
  - Low RPM.
  - A sick passenger.
  - Flight control failure.
  - Instrument failure.
  - System failure.
- In the event of a rejected takeoff:
  - Reduce the power to idle.
  - Apply breaks and aerodynamical breaking.
  - Exit the runway if able.
  - Contact ATC.

**Engine Failure after Takeoff**

- An engine failure after takeoff requires an immediate response.
- Review the takeoff briefing:
  - If sufficient runway is remaining, land on the remaining runway.
  - If no sufficient runway is remaining, make shallow turns, maintain a safe airspeed, and find the best place to land (Ex: if taking off from runway 5 and an engine failure happens, try to turn left to land on 27L).
  - If there is sufficient altitude (above 1,000ft), turn back around to the airport and attempt an engine restart.

        - Review the engine power loss in flight checklist.
- Contact ATC.

**No Flap Approach**
- Ask the student why we use flaps on approach.
    - Lower stall speed (increase in camber, increase in lift coefficient) (VSO vs VS1).
    - Increase in lift.
    - Increase in induced drag.
    - Steeper angle of descent without increasing airspeed.
- Flying a no flap approach:
    - Higher nose attitude.
    - Losing altitude will be more difficult.
    - Wider traffic pattern.
    - Sight picture will be different (highlight this during the flight).
    - The airplane will tend to float more during the round out and flare phases due to the lack of induced drag.
    - A longer ground roll will be experienced.
- When do we do a no flap approach:
    - Flap malfunction.
    - Very windy/gusty conditions.
    - Flying into a busy airport (max forward speed).

**Power Off Approach**
- In the event of an engine failure in the traffic pattern or in close proximity of an airport, a power off approach will be required.
- Contact ATC when able.
    - Pitch for Vglide.
    - Highlight how the winds will drastically affect when to turn base and final.
    - Extending the flaps to 10 degrees helps maintaining a stable traffic pattern, even with an engine failure.
    - Flaps 25 should not be extended until the runway can be made for sure.
    - Flaps 40 should only be used if coming in too high as this drastically increases drag.
    - Once on the runway, turn on the nearest taxiway.

**Go/No-Go Decision**
- The student must make a go/no-go decision depending on:
    - Personal minimums.
    - FIT minimums.
    - Aircraft.
    - Airspace.
- Review the students planning with a focus on:
    - Weather, weight and balance, performance, flight planning, and fuel calculations.
    - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
    - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices,

    - life jackets, fly away kit, survival kit, etc.)
    - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

**Lesson Content (Dual)Checklist Procedures**

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform all briefings.

**Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

**Operation of the Communications and Navigation Units**

- Ensure that the student understands how to properly use COMS and NAV units.
- Refer to points of emphasis from lesson 113.

**Radio Communication Procedures and ATC Phraseology**

- Students should understand and demonstrate proper communication procedures and use of ATC phraseology.
- Refer to points of emphasis from lesson 113.

**Visual Scanning and Collision Avoidance**

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

**Review Previous Exercises as Required**

- Review any areas the student might need help with by utilizing the pertinent lesson plans.

**Ground Reference Maneuvers**

- <u>Student says, student does.</u>
    - Verify correct procedure.
- Points of Emphasis.
    - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
    - Bank variation.
    - Choosing an adequate and safe visual reference point.
    - Provide feedback as needed throughout the maneuver.

**Normal and Crosswind Takeoff**

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

**Abnormalities During Takeoff**

- Rejected Takeoff
  - Ask ATC for a short delay on the runway.
  - As the student adds full power and gains airspeed, bring the throttle to idle to simulate an engine failure.
    - Ensure the student performs the correct procedures.
- Engine Failure After Takeoff
  - Simulate an engine failure on takeoff.
    - Bring the power to idle momentarily.
    - At this point, the student should talk through what they would do while the instructor applies power again and continues with the climb out.

**Traffic Pattern**

- The student will be in charge of all the radio calls with tower/ground or advisory frequency.
- Traffic patterns are to be completed by the student with minimal instructor guidance.
- The student should be able to perform normal, crosswind, and rejected takeoffs and landings with little instructor guidance.

**Abnormalities During Approach and Landing**

- No Flap Approach and Landing
  - On one of the traffic patterns, have the student conduct a no flap approach and landing.
  - The speeds on the base and final legs should be the same.
  - Ensure that the student understands that a wider traffic pattern may be required.
  - The student should understand that excessive floating may happen while rolling out and flaring.
- Power Off Approach and Landing
  - While on the downwind, bring the power to idle.
  - Ensure that the student pitches for 73 knots.
  - Guide the student through when to turn base/final depending on the wind (if needed).
  - The student must be able to land on the first third of the runway.

**Forward Slip to a Landing**

- On one of the traffic patterns, have the student purposely come in high.
- Guide the student through conducting a forward slip.
  - Power should be at idle.
  - Pitch for 63 knots.
  - Ailerons into the wind.
  - Rudder opposite.

**Go-Arounds or Rejected Landings**

- Have the student conduct a go-around during one of the traffic patterns.

- Ensure that the correct procedure is conducted.

**Normal and Crosswind Landing**
- The student should conduct the landing.
- By this lesson, the student should be able to conduct a safe landing with no instructor guidance.

**Postflight Procedures**
- Ensure that the student conducts a thorough postflight inspection.

**Post-Brief**
- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 121.
- Assign homework:

**Completion Standards**
This lesson is complete when the applicant can:
1. Demonstrate proper sequential checklist procedures for all normal and emergency procedures without instructor guidance.
2. Conduct rejected takeoffs without instructor guidance.
3. During ground reference maneuvers, with minimal instructor guidance:
    a. Fly a predetermined ground track using specific ground references.
    b. Understand the effects of wind and correct for wind drift.
    c. Maintain altitude ±150 ft, and airspeed ±10 knots, and a maximum bank of 45°
4. Maintain the recommended traffic pattern altitude ±150 ft, recommended airspeed ±10 knots and correct for wind drift without instructor guidance.
5. Conduct no flap, glide approach and forward or side slips to a landing with instructor guidance.
6. Perform normal and crosswind takeoff and landings with minimal instructor guidance.